Cairo University
Faculty of Computers and Artificial Intelligence

Object-Oriented Programming (OOP)

Assignment 2

Under the supervision of: Dr. Mohamed El-Ramly

Team Members

#	Name	ID
1	Youssef Tamer Mohamed	20242405
2	Ali Mohamed Salah	20242221
3	Amir Eltaher	20242055
4	Hussien Ezzat	20242106

Team Contributions

Member	Contribution
Youssef Tamer Mohamed	• Wrote the Machine class (Machine.h / Machine.cpp): loadProgram() and the fetch-execute loop in run(). • Wrote main.cpp: hexadecimal input, validation, and program execution. • Built VoleMachine.exe and ran the final tests.
Ali Mohamed Salah	• Wrote the Register class (Register.h / Register.cpp). • Wrote the Memory class (Memory.h / Memory.cpp) with 256 cells and range checks. • Wrote the CPU header file (CPU.h) and the constructor, getters, and setters.
Amir Eltaher	• Implemented the LOAD, STORE, and MOVE instructions (opcodes 1, 2, 3, 4). • Implemented the JUMP and HALT instructions (opcodes 11, 12). • Handled printing a character when storing to address 0xFF.
Hussien Ezzat	• Implemented the integer ADD, OR, AND, XOR, and ROTATE instructions (opcodes 5, 7, 8, 9, 10). • Implemented floating-point ADD (opcode 6) with decodeFloat8() and encodeFloat8(). • Tested the program with sample instruction sets.